

Question 1.1. Let's run a hypothesis test using confidence intervals to see if there is a linear relationship between egg weight and bird weight. Define the null and alternative hypotheses that will allow you to conduct this test. **(8 points)**

Note: Please write your answer **in the cell below** in the following format: - **Null Hypothesis:** - **Alternative Hypothesis:**

Hint: Reminder your null hypothesis should be about the relationship in the population.

Null Hypothesis: There is no statistically significant relationship between egg weight and bird weight.

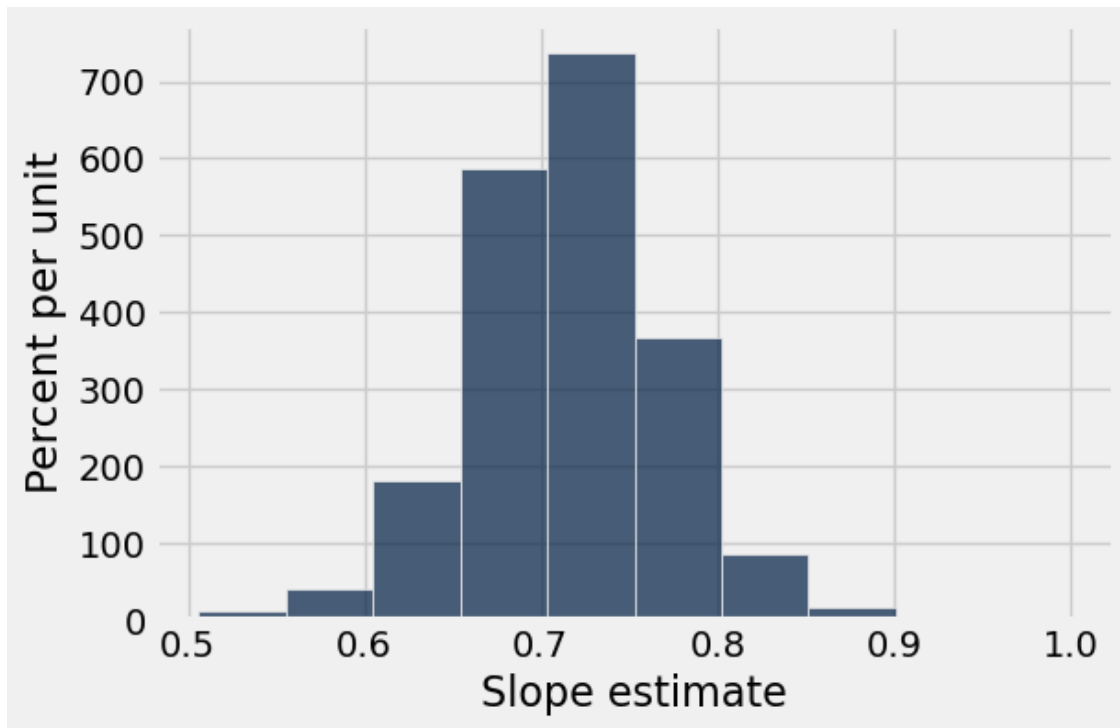
Alternative Hypothesis: There is a statistically significant relationship between egg weight and bird weight, which has a positive correlation.

Question 1.4. Create an array called `resampled_slopes` that contains the slope of the best fit line for 1000 bootstrap resamples of `birds`. Plot the distribution of these slopes. (8 points)

```
In [11]: resampled_slopes = make_array()

for i in np.arange(1000):
    birds_bootstrap = birds.sample()
    bootstrap_line = fit_line(birds_bootstrap, "Egg Weight", "Bird Weight")
    bootstrap_slope = bootstrap_line[0]
    resampled_slopes = np.append(resampled_slopes, bootstrap_slope)

# DO NOT CHANGE THIS LINE
Table().with_column("Slope estimate", resampled_slopes).hist()
```



```
In [12]: grader.check("q1_4")
```

```
Out[12]: q1_4 results: All test cases passed!
```


Question 1.6. Based on your confidence interval, would you accept or reject the null hypothesis that the true slope is 0? Why? What p-value cutoff are you using? **(8 points)**

Hint: Read the introduction of this homework!

Given that the 95% confidence interval means that we are using a 5% p-value cutoff, we must reject the null hypothesis because the slope within the 95% confidence interval does not contain 0 and instead contains intervals that are greater than 0.

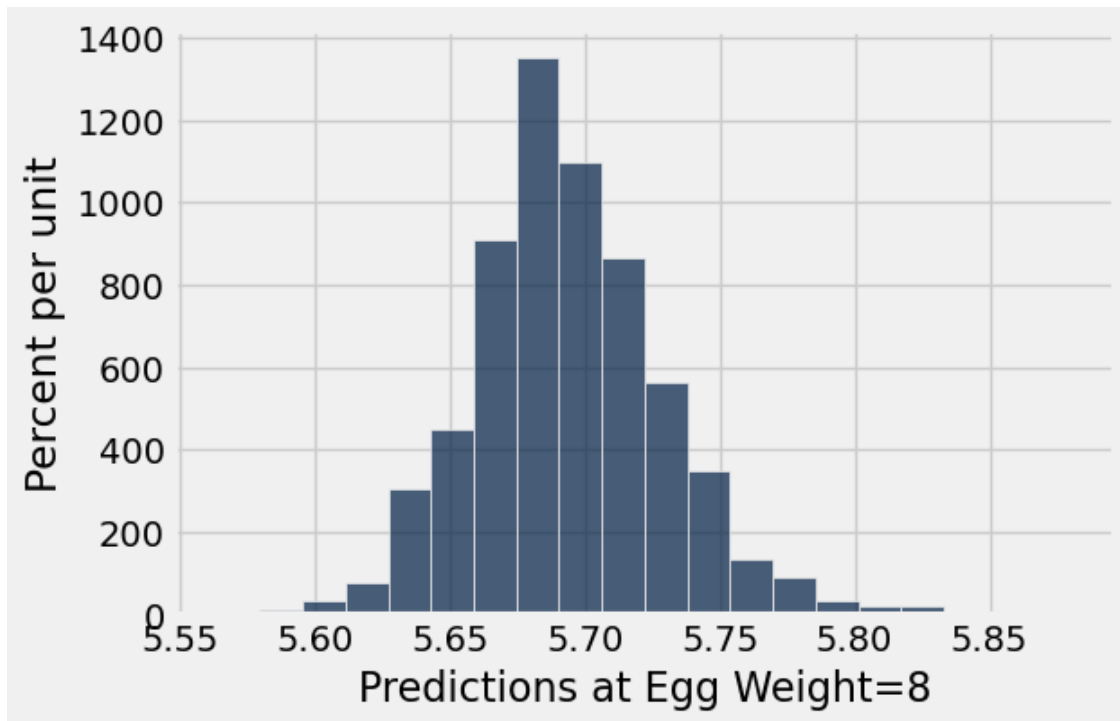
Question 1.7. Using your results from previous questions, **provide an estimated range for the true slope**. How confident are you that the true slope falls within this range? **(8 points)**

By using the bootstraps and the 95% confidence interval, the true slope falls in the interval between 0.7 and 0.73 since that is where the peak of the bell curve of the distribution would be. A rough estimate for the true slope would be about 0.715.

Question 2.4. Create an array called `predictions_for_eight` that contains the predicted bird weights based on an egg of weight 8 grams for each regression line in `regression_lines`. (6 points)

```
In [23]: predictions_for_eight = regression_lines.column("Slope")* 8 + regression_lines.column("Intercept")

# This will make a histogram of your predictions:
table_of_predictions = Table().with_column('Predictions at Egg Weight=8', predictions_for_eight)
table_of_predictions.hist('Predictions at Egg Weight=8', bins=20)
```



```
In [24]: grader.check("q2_4")
```

```
Out[24]: q2_4 results: All test cases passed!
```

